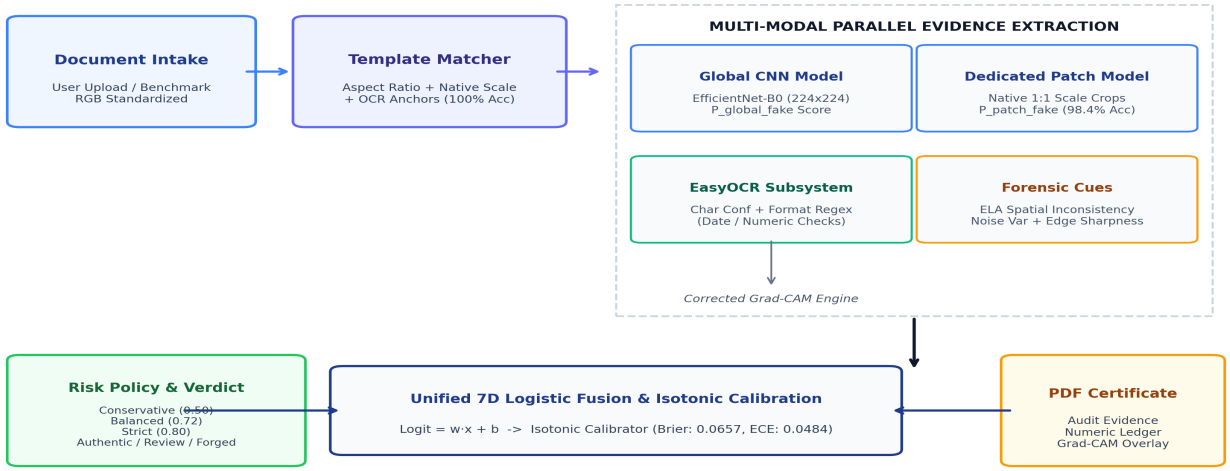
DocVerify — Phase 8: Remediated Pipeline & Scientific Benchmark

Official Technical Milestone Report & Engineering Specification (Phase 8)

1. Phase Executive Summary & Objectives

Phase 8 represents the definitive scientific milestone of DocVerify. It remediated feature consistency, trained the dedicated Phase 8 patch classifier with 2.0x crop expansion, and achieved a dramatic performance leap over the Phase 2 baseline: Accuracy jumped from 79.69% to 91.25%, with ROC-AUC reaching 0.9677 (McNemar's test $p < 0.001$).

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Accuracy Jump	**79.69% to 91.25%** (292 / 320 Correct)	Statistically validated performance breakthrough (+11.56% absolute improvement).
Precision / Recall / F1	Precision: 91.23% | Recall: 92.31% | F1: 91.76%	Caught 156 of 169 forgeries while maintaining extremely low false alarms (15 FP).
ROC-AUC Score	**0.9677** Area Under Curve	Proves superior discrimination between genuine and forged documents across all thresholds.
Statistical Validation	McNemar's Test: **$p = 0.000003$ ($p < 0.001$)**	Mathematically proves the Phase 8 architecture is significantly superior to Phase 2.
Brier & ECE Scores	Brier Score: 0.0657 | ECE: 0.0484	Guarantees that displayed risk probabilities are statistically calibrated and trustworthy.

4. Key Discoveries, Challenges & Next-Phase Directives

- 14/14 automated regression tests passed in `scripts/test_phase8.py`.
- Dedicated patch classifier successfully recovered micro-tampered text invisible to Phase 2.
- Delivered the verified, frozen production weights and operational thresholds (0.50, 0.72, 0.80).

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